

Mobile Computing

Lecture 1

Introduction to Mobile Computing

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Course Objectives

- Understand fundamental concepts in mobile wireless systems.
- Understand fundamental concepts, challenges, and demands of mobile computing.
- Cover systems and standards issues including: Wireless LANs, Mobile IP, ad-hoc networks, sensor networks.
- Cover issues associated with handheld portable devices.
- Learn about new applications that can exploit mobility and location information.

CSE 4721 Mobile Computing

Topics to be covered

- Give a brief overview of fundamental concepts in mobile computing.
- Give a brief overview of fundamental concepts in mobile wireless systems.
- Wireless LANs.
- Mobile IP
- Ad-hoc networks.
- Sensor networks.
- Issues with small handheld devices.
- New applications that can exploit mobility and location information.

Grading Scheme

- Year Work (20)
- Practical/Oral (20)
- Final Exam (60)

كود المقرر	اسم المقرر	عدد الساعات الأسبوعية			توزيع الدرجات			عدد ساعات الامتحان التحريري	مجموع درجات المقرر	الساعات المعتمدة
		محاضرة	تطبيقات	مجموع	أعمال سنة	عملية	تحريري			
حاس ٤٧٢٠	النمذجة والمحاكاة	3	-	1	4	20	20	60	100	3
حاس ٤٧٢١	الحوسبة المتنقلة	3	-	1	4	20	20	60	100	3
عام ٤٧٢٢	دراسات بيئية	2	-	-	2	15	-	35	50	2
-	مقرر اختياري (٤)	3	-	1	4	30	30	90	150	3
-	مقرر اختياري (٥)	3	-	1	4	30	30	90	150	3
حاس ٤٧٣٩	مشروع التخرج	-	-	4	4	100	-	-	200	2
إجمالي عدد الساعات الأسبوعية		14	-	8	22	عدد الساعات المعتمدة			16	
					المجموع الكلي للدرجات			1500		

Introduction to Mobile Computing

What is Mobile Computing?

Mobile Computing is a term that describes technologies that enable users to access network services anywhere and at any time.

What are the enabling technologies for mobile computing?

- Cellular communication.
- Wireless LANs.
- Satellite services.
- Etc.

Introduction to Mobile Computing



Main characteristics

- Users are no longer required to maintain a fixed position in the network.
- Mobile computers will be equipped with a wireless connection to the fixed part of the network.
- Almost unrestricted mobility.
- Increased portability.

Introduction to Mobile Computing



Types of communication devices

- **Fixed and wired** (e.g. desktop computers)
- **Mobile and wired** (e.g. laptops reconnecting to company's network via the telephone network and a modem)
- **Fixed and wireless** avoid damage by installing wires (e.g. in historical buildings)
- **Mobile and wireless** User can roam between wireless networks. (e.g. 4G, 5G networks)

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Applications of Mobile Computing

- **Vehicles:** News, weather reports, and road conditions are received via digital audio broadcasting (DAB). Driven cars in the same area build an ad-hoc network for fast exchange of urgent information in the time of emergency (e.g, accident)
- **Emergencies:** Ambulances could send vital information about injured people to the hospital. In the case of natural disasters, wireless networks may be the only means of communication.
- **Business:** Gives a competitive edge for businesses by providing latest information during critical presentations to important customers. A salesman with a laptop could be viewed as carrying a mobile office.

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Applications of Mobile Computing

- **Credit Card Verification:** Point of Sale (POS) terminals communicates securely over cellular channels using a mobile computer unit to verify customer transaction with the bank's central computer.
- **Replacement of Wired Networks:** Sometimes is too expensive and impractical to use wires. (e.g. remote sensors, for weather forecast).
- **Infotainment:** provide location-based up-to-date information. (e.g. travel guide, ad-hoc gaming networks)

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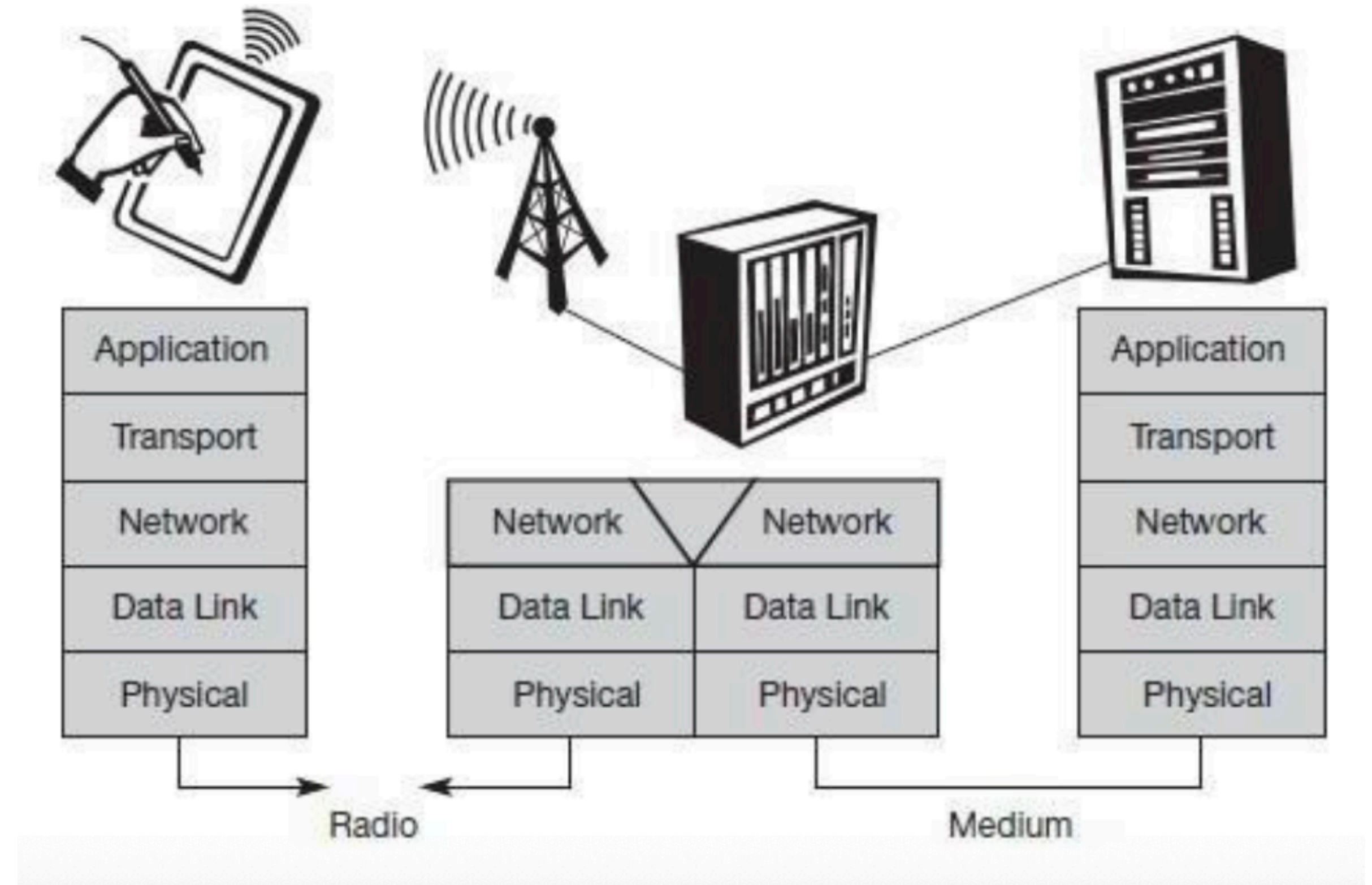
Limitations of Mobile Computing

- **Resource constraints:** Limited battery capacity, processing power, and memory.
- **Interference:** Unavoidable due to radio transmission. Hence, high loss rates for transmitted data and higher bit error rates.
- **Bandwidth:** Relatively lower than desktop systems. Part of the problem is due to communication protocols overhead.
- **Dynamic changes in communication environment:** Signal power instability. Hence, disconnections and link delays.
- **Network issues:** Connection stability.
- **Interoperability issues:** supporting different protocol standards.
- **Security constraints:**
 - Portable devices are more prone to getting lost or stolen.
 - The ease of eavesdropping on radio communications.
 - Wireless communications should employ efficient encryption, authentications, and other security measures.

Introduction to Mobile Computing

A simplified reference model

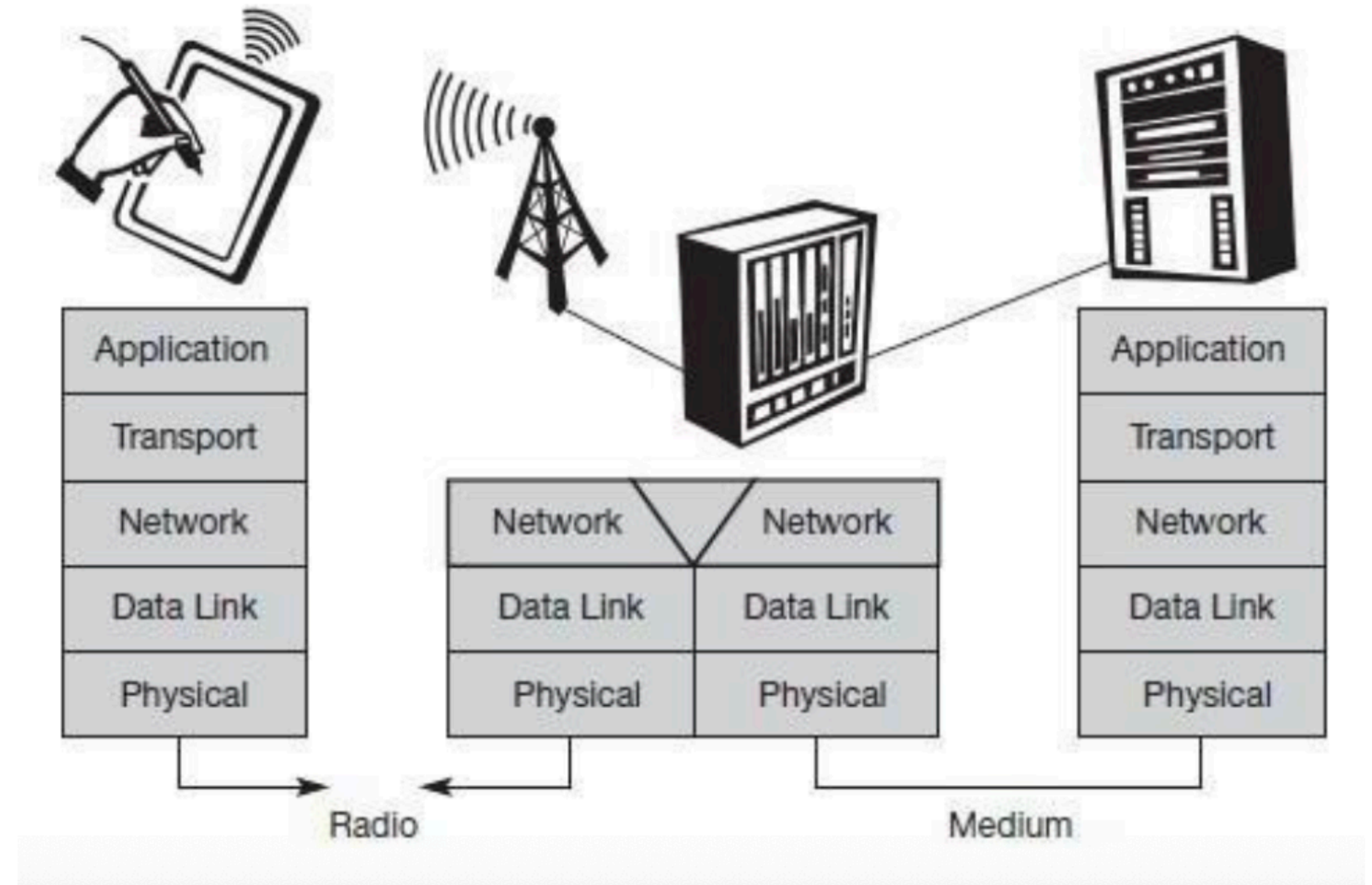
- **End-systems** need full protocol stack.
- **Intermediate systems** do not necessarily need the full stack.



Introduction to Mobile Computing

A simplified reference model

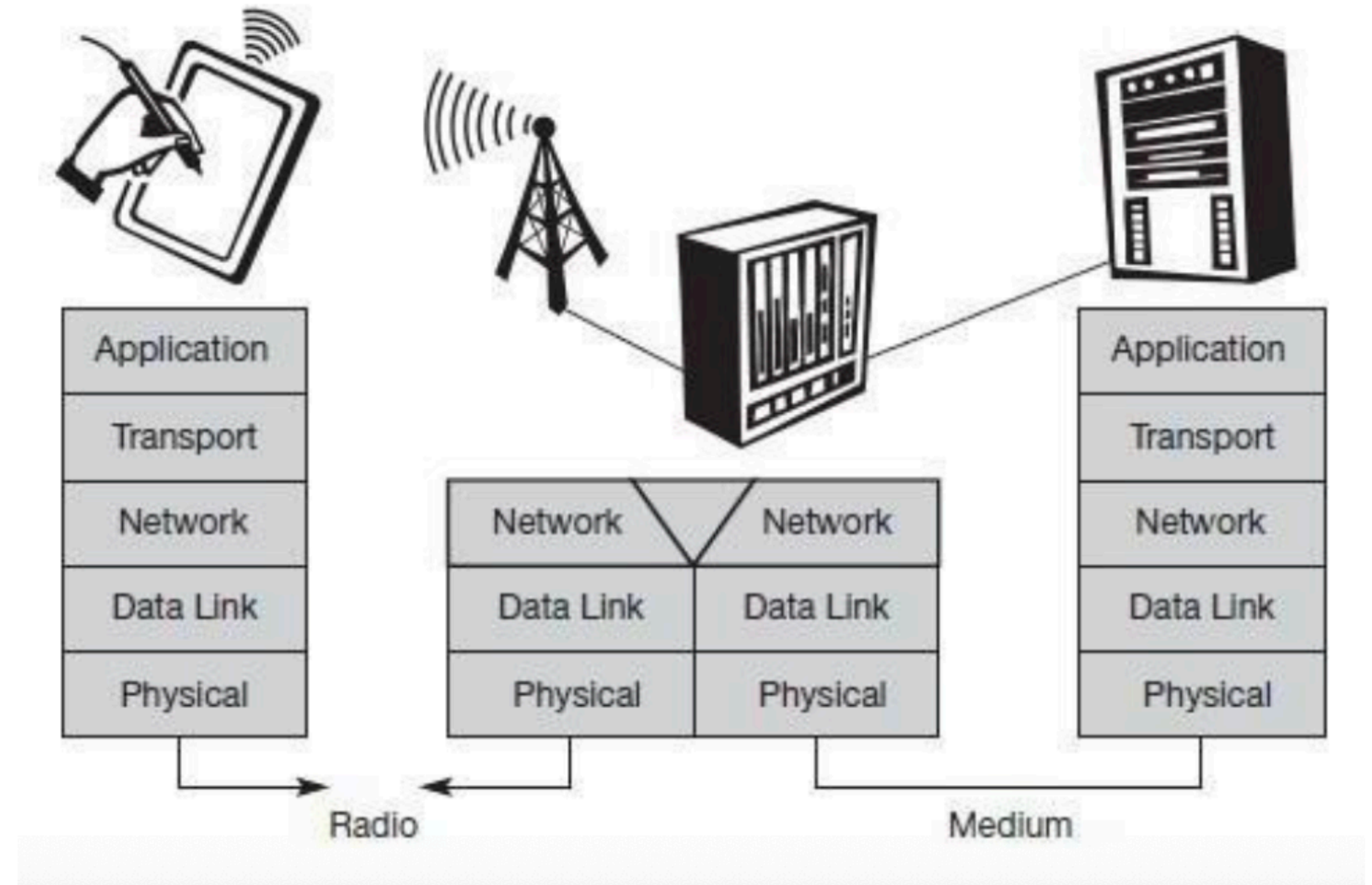
- **Physical layer:** conversion of a stream of bits into transmission signals at the sender. Also, conversion of received signals back into stream of bits at the receiving end.
- For wireless communication, the physical layer is in charge of:
 - Frequency selection
 - Carrier frequency generation
 - Signal detection within a noisy environment.
 - Modulation of data onto a carrier frequency.
 - Encryption (sometimes)



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A simplified reference model

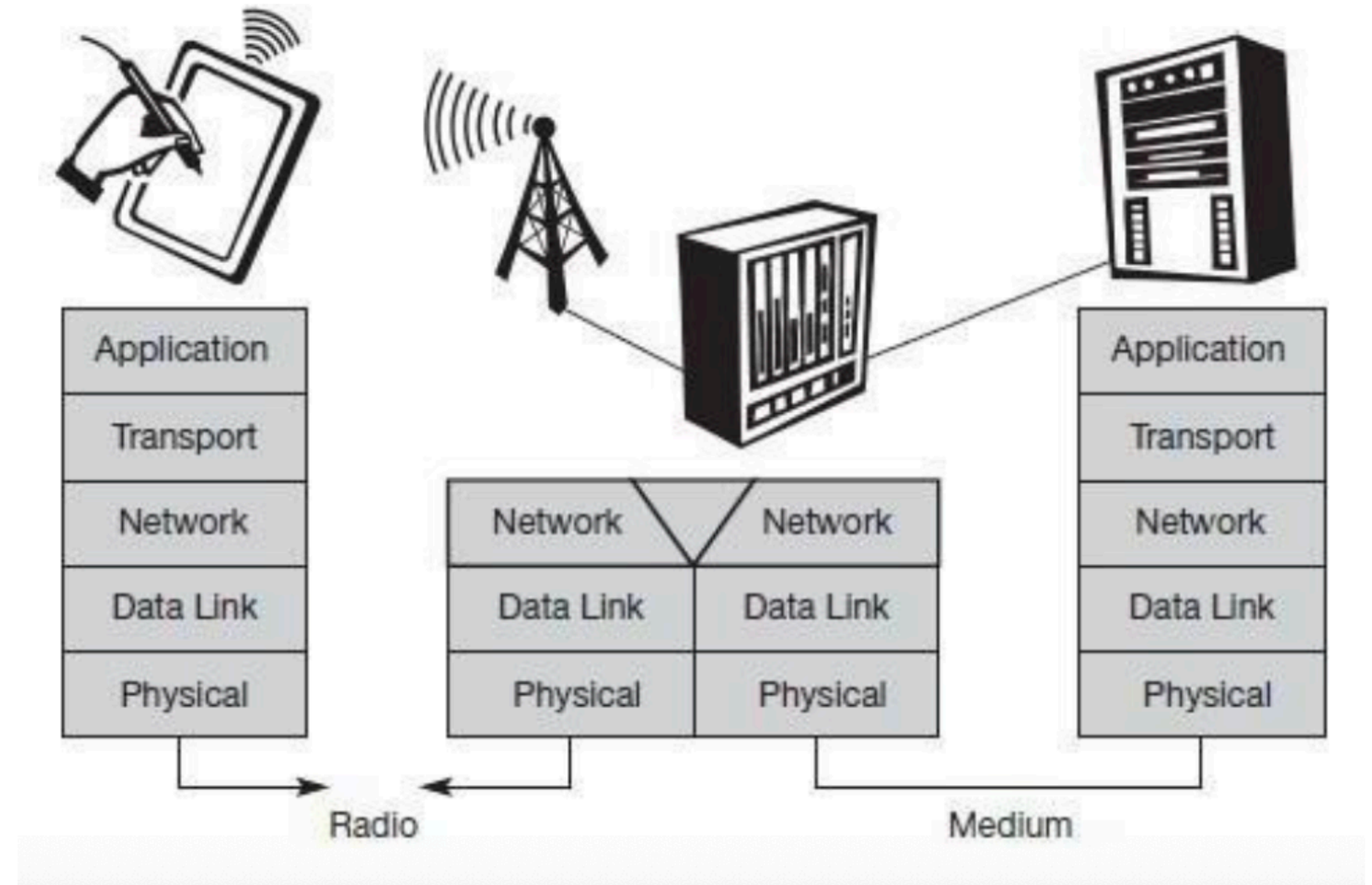
- **Data link layer** is responsible for reliable point-to-point or point-to-multipoint connectivity:
 - Medium access.
 - Multiplexing of different data streams.
 - Error correction
 - Synchronization (Data frame detection)



Introduction to Mobile Computing

A simplified reference model

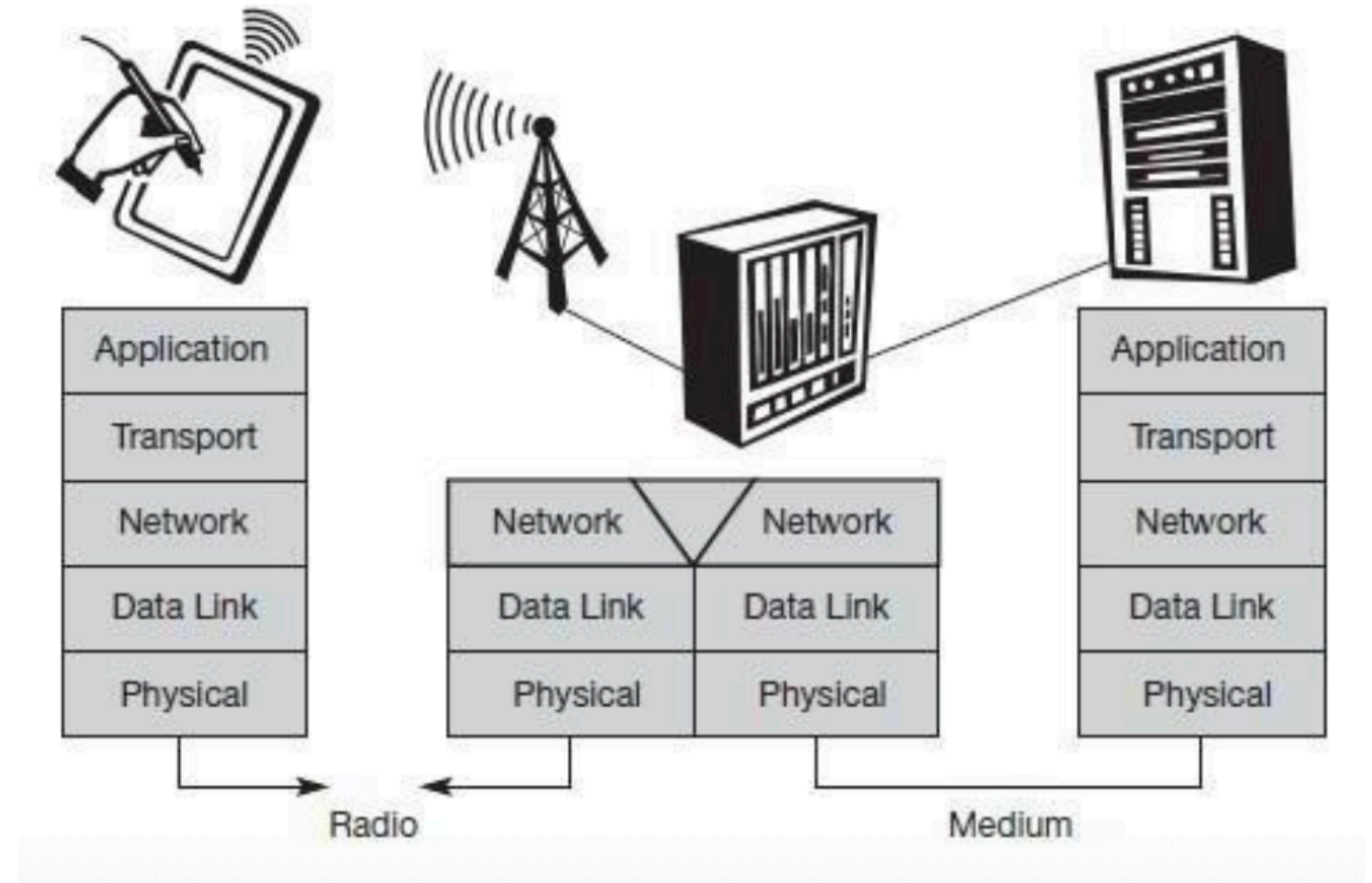
- **Network layer** is responsible for routing packets through a network to establish a connection between two devices over one or more intermediate nodes.
- Network layer tasks
 - Addressing
 - Routing
 - Device location
 - Handover



Introduction to Mobile Computing

A simplified reference model

- **Transport layer** is responsible for establishing an end-to-end connection.
- **Application layer** performs the following functions:
 - Service location
 - Multimedia support
 - Adaptive applications
 - Wireless access to the Internet



Mobile Ad-Hoc Networks (MANETs)

Introduction

- A MANET is an autonomous collection of mobile users that communicate over relatively bandwidth constrained wireless links.
- MANETs are wireless networks that has no fixed infrastructure.
- MANETs are characterized by dynamic topologies that changes rapidly due to nodes mobility.
- MANETs are decentralized networks.
- Each node may act as both a host and a router.
- So, some nodes forward packets to other nodes in the network.

